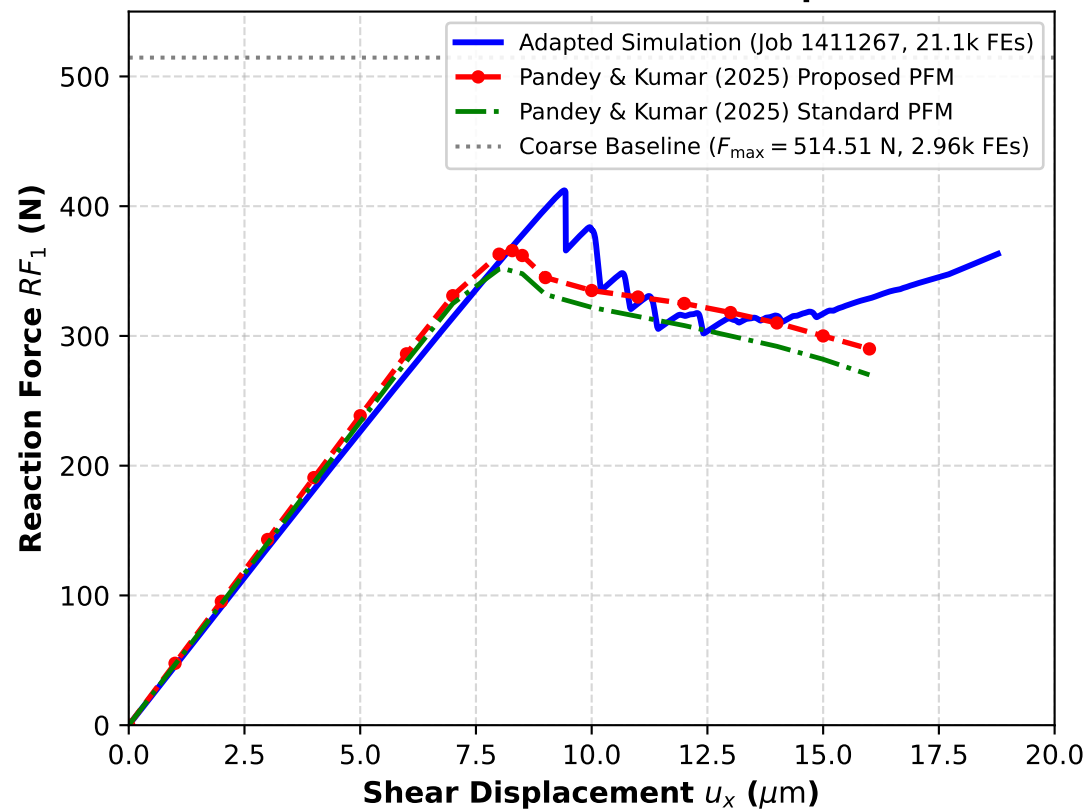
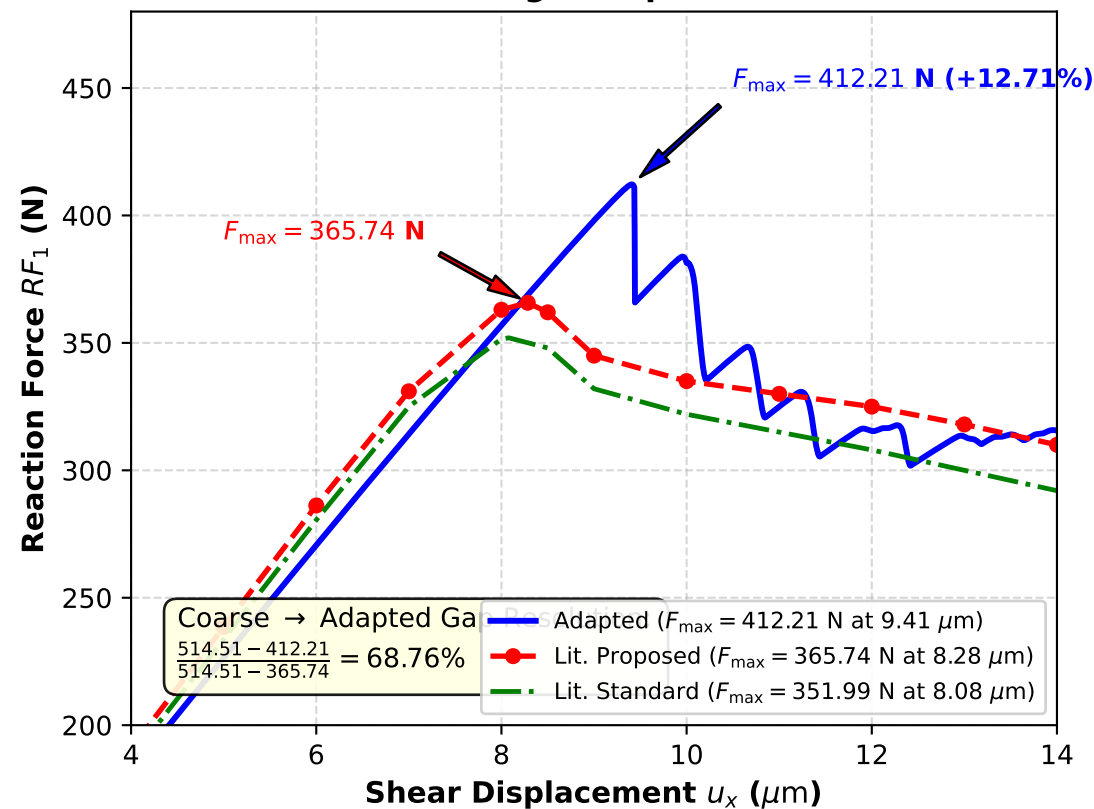
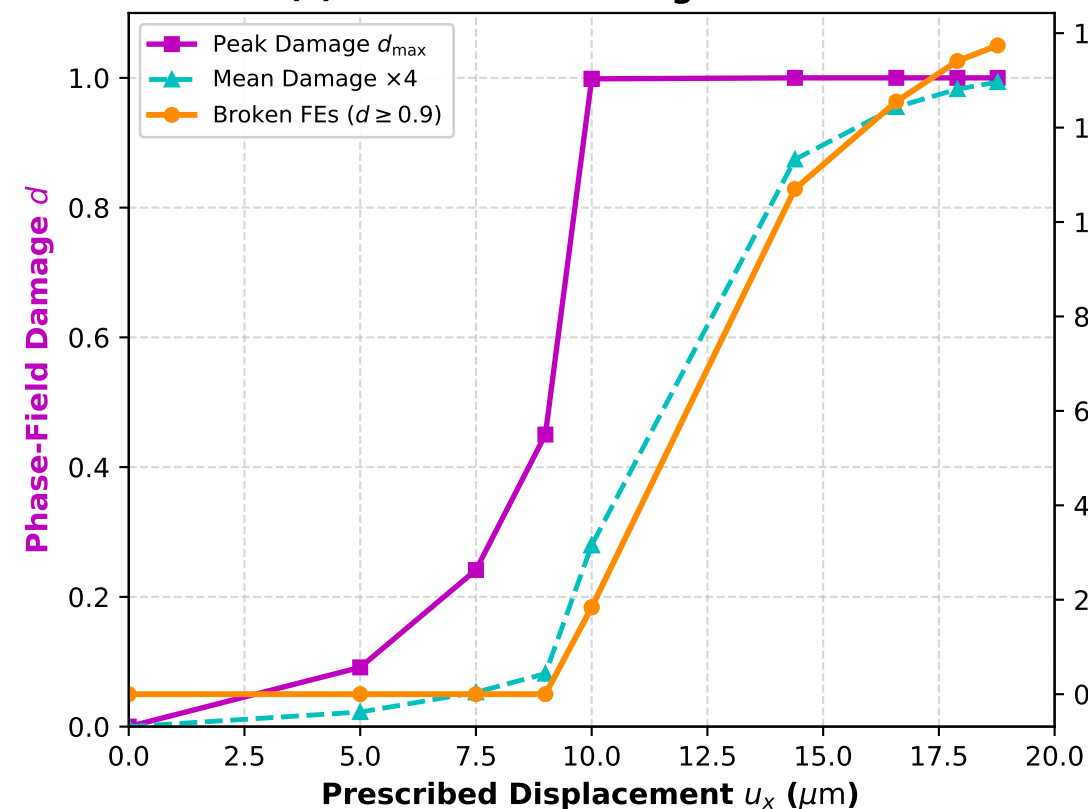


(a) Mode-II Reaction Force vs. Displacement**(b) Peak Softening & Gap Resolution Audit****(c) Phase-Field Damage Evolution****(d) Crack Front Penetration & Intact Ligament**